

Multiobjective Optimization

A-priori Methods: Concepts and Examples

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Key Concept:

"Generate the noninferior set first, then let decision makers choose."

Methods Covered:

- Weighting Method
- Constraint Method
- NISE Method
- Multiobjective Simplex

Today: A-priori Methods - Preferences are articulated **before** solution

What Are A-priori Methods?

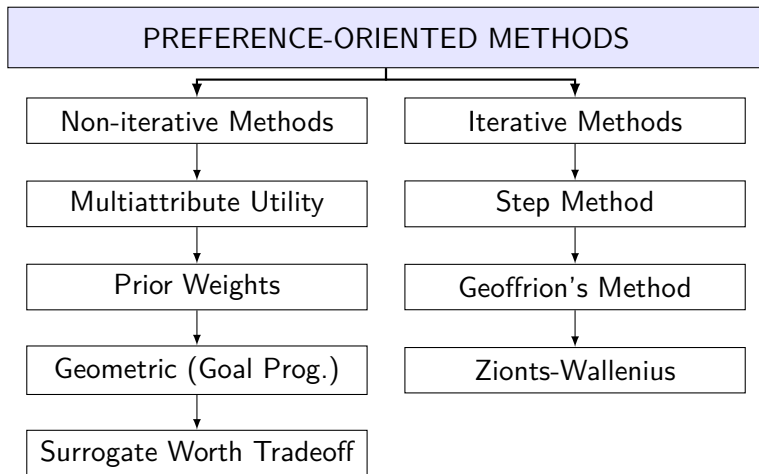
Definition:

*Methods that require decision makers to articulate their preferences **before** the solution is found.*

Key Characteristics:

- Preferences are made **explicit**
- Information flows **top-down** (DM $\rightarrow$ Analyst)
- Goal is to identify **best-compromise solution** directly
- Generally **less computationally intensive** than generating methods

Types of Preference-Oriented Techniques



How Preferences Are Expressed

Decision Makers Articulate:

Type of Information	Example
Weights	$w_1 = 0.6, w_2 = 0.4$
Goals/Targets	$Z_1 \geq 100, Z_2 \geq 50$
Utility Functions	$U(Z_1, Z_2) = Z_1 \cdot Z_2$
Tradeoffs	"I'd sacrifice 2 units of Z_2 for 1 of Z_1 "
Lexicographic Ordering	$Z_1 \gg Z_2 \gg Z_3$

Challenge: Extracting reliable preference information

Definition: A function $U[Z_1(\mathbf{x}), Z_2(\mathbf{x}), \dots, Z_p(\mathbf{x})]$ that assigns a single utility value to each combination of objectives.

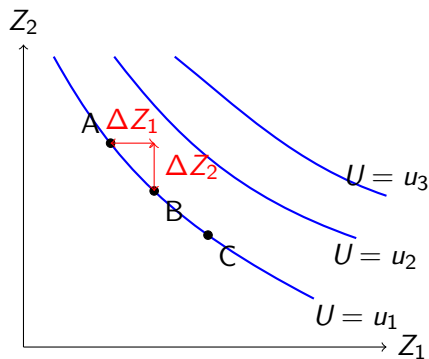
Properties (for maximization):

$$\frac{\partial U}{\partial Z_k} > 0 \quad \forall k \quad (\text{More is better})$$

$$\frac{\partial^2 U}{\partial Z_k^2} < 0 \quad \forall k \quad (\text{Diminishing returns})$$

Result: Complete ordering of all alternatives

Graphical Representation



MRS: Amount of Z_2 sacrificed for one unit of Z_1

Key Concept for Utility

Definition:

$$\text{MRS}_{ik} = - \frac{\partial Z_k}{\partial Z_i} \Big|_{U=\text{constant}}$$

Interpretation:

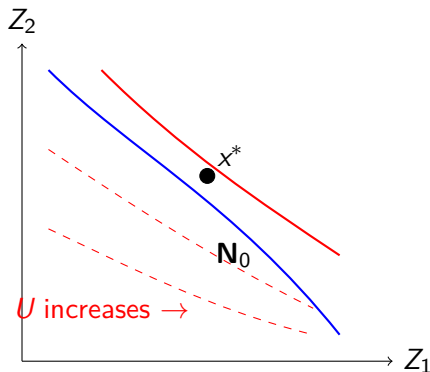
- Amount of objective k sacrificed for 1 unit of objective i
- Varies along indifference curves (diminishing MRS)
- Represents **desirable** tradeoff

Note: Compare with **MRT** (feasible tradeoff from noninferior set)

Using Utility Functions

Condition for Best-Compromise:

$$MRS_{12} = MRT_{12}$$



At x^* : Utility is maximized on N_0

General Form:

$$U(Z_1, Z_2, \dots, Z_p) = \sum_{k=1}^p U_k(Z_k)$$

Assumptions:

- Utility independence among objectives
- No interaction effects

Advantages:

- Easier to estimate
- Decision makers can consider each objective separately

Disadvantage:

- May not capture complex preferences

Finding Best-Compromise with Utility

Problem:

$$\begin{aligned} &\text{maximize } U[Z_1(\mathbf{x}), Z_2(\mathbf{x})] \\ &\text{s.t. } \mathbf{x} \in \mathbf{F}_d \end{aligned}$$

Approach (for two objectives):

- 1 Start at one endpoint of $\mathbf{N}_0$
- 2 Parametrically vary weights in weighted problem
- 3 Compare utility at successive solutions
- 4 When utility decreases, best-compromise lies between last two points
- 5 Search along line segment for maximum

Given:

$$U(Z_1, Z_2) = Z_1 \cdot Z_2$$

Noninferior points from sample problem:

Point	Z_1	Z_2	U
P_0	-3	15	-45
P_1	12	12	144
P_2	26	2	52

Result: Best-compromise lies between P_1 and P_2

Refining the Solution

Between P_1 and P_2 :

$$Z_1 = 12 + t(26 - 12) = 12 + 14t$$

$$Z_2 = 12 + t(2 - 12) = 12 - 10t$$

Maximize:

$$\begin{aligned} U &= (12 + 14t)(12 - 10t) \\ &= 144 - 120t + 168t - 140t^2 \\ &= 144 + 48t - 140t^2 \end{aligned}$$

Optimal: $t = 0.83$

Result: $(Z_1^*, Z_2^*) = (14.4, 10.3)$

Simplest A-priori Method

Concept: Decision makers assign weights to objectives **before** analysis.

Formulation:

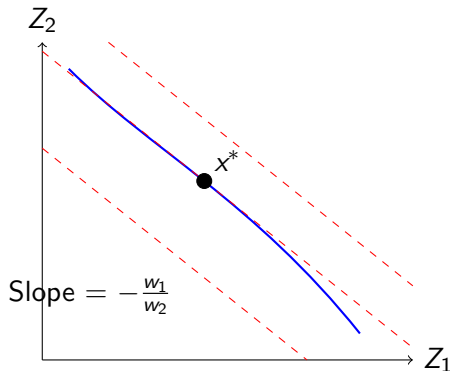
$$\text{maximize } Z(\mathbf{x}, \mathbf{w}) = \sum_{k=1}^p w_k Z_k(\mathbf{x})$$

Interpretation:

- Weights represent **constant marginal rates of substitution**
- Indifference curves are **linear**

Note: Only **relative weights** matter (w_i/w_j)

Linear Indifference Curves



Limitation: Constant MRS (w_1/w_2)

Advantages:

- Simple and computationally efficient
- Straightforward to explain
- Allows sensitivity analysis

Disadvantages:

- Constant MRS may be unrealistic
- Decision makers may not be able to articulate weights
- Zeleny (1974): "Precision of weighting does not reflect fuzziness of multiobjective decision making"

Recommendation: Extensive sensitivity analysis on weights

Concept: A separate policy committee sets national parameters (weights) for all projects.

Motivation:

- Avoid bureaucratic bias
- Consistent policy across agencies
- Decision makers focus on policy, not design

Challenge:

- Objectives and measures vary by project
- Different weights may be appropriate in different contexts

A-priori Methods:

- Preferences articulated **before** solution
- Goal: Identify **best-compromise** directly
- Less computationally intensive
- Require identifiable, accessible decision makers

Key Tools Covered:

- 1 Multiattribute Utility Functions - Complete preference representation
- 2 Prior Weights - Simplest approach (linear indifference curves)

Next: More advanced a-priori methods

Topics:

- ① Geometrical Definitions of Best
 - Minimizing distance from ideal
 - Goal Programming
- ② Surrogate Worth Tradeoff Method
- ③ Iterative Techniques
 - Step Method
 - Geoffrion's Interactive Method

Key Question: How do we make preference articulation practical?



COHON, J. L. *Multiobjective Programming and Planning*. Dover Publications, 2013.